

Oral Surgery: Orthodontic Related Procedures

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 [Instructions for Use](#)

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Related Dental Policy

- [Medically Necessary Orthodontic Treatment](#)

Related Medical Policy

- [Orthognathic \(Jaw\) Surgery](#)

Coverage Rationale

Surgical Placement of Temporary Anchorage Device (Not Related to Distraction Osteogenesis or Orthognathic Surgery)

The surgical placement of temporary [Anchorage](#) devices are used in conjunction with orthodontic treatment and are indicated for individuals age 12 and over for the following:

- [Intrusion](#) of maxillary teeth
- Molar [Distalization](#)
- Canine [Retraction](#) and Intrusion Retraction mechanics
- Correction of anterior [Open Bite](#) and deep [Overbite](#)
- Correction of a canted [Occlusal Plane](#)

The surgical placement of a temporary Anchorage device may not be indicated for individuals with any of the following:

- Known allergy to titanium alloy
- History of heavy tobacco use
- Advanced osteoporosis
- Uncontrolled immune or metabolic bone disorders
- Individuals with an unmanaged medical condition; these conditions include but are not limited to metabolic, cardiovascular, and autoimmune/inflammatory, as well as genetic conditions that affect collagen synthesis
- Individuals taking medications that negatively affects the healing response; these include but are not limited to immunosuppressive agents, corticosteroids, anticoagulants, NSAIDS, and nicotine
- Poor oral hygiene
- Poor quality cortical bone density and volume
- [Ankylosed](#) teeth

Surgical Access of Unerupted Tooth

Surgical access of unerupted tooth is indicated for the following:

- When a normally developing permanent tooth is unable to erupt into a functional position
- For labially impacted teeth if there will be 2-3 mm of gingival cuff present after eruption

Surgical access of unerupted tooth may not be indicated for the following:

- For supernumerary teeth and third molars
- When surgical access of impacted tooth would threaten vital structures

- Individuals with an unmanaged medical condition; these conditions include but are not limited to metabolic, cardiovascular, and autoimmune/inflammatory, as well as genetic conditions that affect collagen synthesis
- Individuals taking medications that negatively affects the healing response; these include but are not limited to immunosuppressive agents, corticosteroids, anticoagulants, NSAIDS, and nicotine

Placement of Device to Facilitate Eruption of Impacted Tooth

This is the placement of an orthodontic bracket, band, or other device and attached with a chain, on an unerupted tooth, after surgical exposure, to aid in its eruption. This procedure is done following the surgical access of an unerupted tooth.

Corticotomy

Corticotomy is unproven due to insufficient evidence of efficacy and/or safety.

Mobilization of Erupted or Malpositioned Tooth to Aid Eruption

Mobilization of erupted or malpositioned tooth to aid eruption is indicated for the treatment of ankylosed permanent teeth.

Definitions

Anchorage: Resistance to force. Anchorage may come from any of the following sources: intraoral (teeth, bone, soft tissue, implants), or extraoral (cervical, occipital, cranial). (AAO)

Ankylosis: Abnormal immobility, union, or fusion. It may occur between two bones at their articulation (e.g., TMJ) or between teeth and alveolar bone. (AAO)

Corticotomy: A surgical procedure that intentionally inflicts mechanical damage on the cortical bone to increase bone remodeling, accelerate the repair, and shorten orthodontic treatment time. (Fernández-Ferrer)

Distalization: A common descriptor for the biomechanics involved in moving maxillary first and second molars distally and into a Class I molar relationship. (AAO)

Intrusion: A translational form of tooth movement directed apically and parallel to the long axis of a tooth. (AAO)

Occlusal Plane: The imaginary surface on which upper and lower teeth meet in occlusion. It is actually a compound curved surface but is commonly approximated by a plane (straight line in the lateral view) based on specific reference points within the dental arches. (AAO)

Open Bite: Lack of tooth contact in an occluding position (also called apertognathia). (AAO)

Orthognathic Surgery: Orthognathic Surgery is the surgical correction of abnormalities of the mandible, maxilla, or both. The underlying abnormality may be present at birth or may become evident as the patient grows and develops or may be the result of traumatic injuries. The severity of these deformities precludes adequate treatment through dental treatment alone. (AAOMS)

Overbite: Vertical overlap of maxillary teeth over mandibular anterior teeth, usually measured perpendicular to the Occlusal Plane. (AAO)

Retraction: Pertaining to desired posteriorly directed, orthodontic or orthopedic displacements of teeth or of bones of the face. (AAO)

Applicable Codes

The following list(s) of procedure and/or diagnosis codes is provided for reference purposes only and may not be all inclusive. Listing of a code in this policy does not imply that the service described by the code is a covered or non-covered health service. Benefit coverage for health services is determined by the member specific benefit plan document and applicable laws that may require coverage for a specific service. The inclusion of a code does not imply any right to reimbursement or guarantee claim payment. Other policies and guidelines may apply.

CDT Code	Description
D7280	Exposure of an unerupted tooth
D7282	Mobilization of erupted or malpositioned tooth to aid eruption
D7283	Placement of device to facilitate eruption of impacted tooth
D7292	Placement of temporary anchorage device (screw retained plate) requiring flap
D7293	Placement of temporary anchorage device requiring flap
D7294	Placement of temporary anchorage device without flap
D7296	Corticotomy – one to three teeth or tooth spaces, per quadrant
D7297	Corticotomy – four or more teeth or tooth spaces, per quadrant
D7298	Removal of temporary anchorage device [screw retained plate], requiring flap
D7299	Removal of temporary anchorage device, requiring flap
D7300	Removal of temporary anchorage device without flap
D7997	Appliance removal (not by dentist who placed appliance), includes removal of archbar

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Description of Services

Temporary Anchorage devices (TADs) are used to improve Anchorage during routine orthodontic therapy and can allow better Anchorage than extraoral headgear which relies on significant patient compliance for success is one that is placed on the bone for a limited amount of time in order to improve Anchorage in one of two ways: (i) by providing support for the teeth of the reactive unit (indirect anchorage) or (ii) by eliminating the need for the reactive unit (direct Anchorage), and is then removed.

Impacted teeth are those that are not expected to erupt into their normal position within the dental arch and may require surgery to expose the tooth and place a bracket, band, or other device on the unerupted tooth after its exposure to aid eruption.

Corticotomy [also known as periodontally accelerated osteogenic orthodontics (PAOO) or surgically assisted osteogenic orthodontics (SAOO)] is a relatively new surgical procedure that involves creating cuts in, or removal of alveolar bone for the purpose of accelerating orthodontic movement, and reducing treatment time.

Pursuant to CA AB2585: While not common in dentistry, nonpharmacological pain management strategies should be encouraged if appropriate.

Clinical Evidence

Temporary Anchorage Devices

Barreneche-Calle et al. (2024) conducted a systematic review and meta-analysis comparing temporary anchorage device-assisted maxillary expansion (bone-borne or hybrid appliances) versus conventional tooth-borne protocols, in individuals with transverse maxillary deficiency associated with edge-to-edge or unilateral/bilateral crossbite who are still growing. Nine clinical trials (n = 377; mean age 13.2 ± 0.6) were included; and across seven studies that reported sex, 154/254 (60%) were female and 100/254 (40%) were male. The primary endpoint was buccal coronal width between first premolars (distance between buccal cusp tips of the right and left first premolars). The results showed measurable dentoalveolar and periodontal changes after expansion, which were greater on the maxillary first premolars in tooth-borne appliances. Meta-analyses were included from two studies (n = 64); patients who used tooth-borne appliances had greater effects of buccal intercoronal width between the premolars with statistically significant differences (Mean difference 2.34; 95% CI: 0.04–4.65 p = 0.05). Conversely, those patients who used bone-borne or hybrid appliances had greater effects of buccal intercoronal width between molars with statistically significant differences (Std Mean difference 0.64; 95% CI: 1.38–0.10; p = 0.09) with a pooled standardized mean difference of 2.34 (95% CI 0.04 to 4.65; p = 0.05). The authors concluded that it is challenging to recommend a specific expansion protocol due to substantial heterogeneity of information reported. The evidence base was limited by the availability of clinical trials and a high risk of bias among the included studies. Long-term stability was not demonstrated as most studies assessed only immediate effects at the end of treatment.

Antoszewska-Smith et al. (2017) conducted a systematic review and meta-analysis to compare the effectiveness of orthodontic miniscrew implants-temporary intraoral skeletal anchorage devices (TISADs)-in anchorage reinforcement during en-masse retraction in relation to conventional methods of anchorage. A search of PubMed, Embase, Cochrane Central Register of Controlled Trials and Web of Science were performed. Relevant articles were assessed for quality according to Cochrane guidelines and the data extracted for statistical analysis. A meta-analysis of raw mean differences concerning anchorage loss, tipping of molars, retraction of incisors, tipping of incisors, and treatment duration was carried out. Fourteen articles that included 616 individuals were selected as meeting the criteria. The quality of the studies was assessed as moderate. Meta-analysis showed that use of TISADs facilitates better anchorage reinforcement compared with conventional methods. On average, TISADs enabled 1.86 mm more anchorage preservation than did conventional methods. The authors concluded that TISADs are more effective than conventional methods of anchorage reinforcement. The average difference of almost 2 mm seems not only statistically but also clinically significant. However, the results should be interpreted with caution because of the moderate quality of the included studies. More high-quality studies on this issue are necessary to enable drawing more reliable conclusions.

Heravi et al (2016) conducted a study evaluating the movement of impacted canines away from the roots of neighboring teeth before full-mouth bracket placement, performed by means of TADs to decrease undesired side effects on adjacent teeth. The study sample consisted of 34 palatally impacted canines, 19 in the experimental group and 15 in the control group. In the experimental group, before placement of brackets, the impacted canine had erupted by means of miniscrews. In the control group, after initiation of comprehensive orthodontics, canine disimpaction was performed by means of a cantilever spring soldered to a palatal bar. At the end of treatment, volume of lateral incisors and canine root resorption were measured and compared by means of a CBCT-derived tridimensional model. Visual Analogue Scale (VAS) score, bleeding on probing (BOP) and gingival index (GI) were recorded. Clinical success rate was also calculated. The volume of root resorption of lateral teeth in the control group was significantly greater than in the experimental group ($p < 0.001$). At the end of treatment, VAS score, GI and BOP were not significantly different between the two groups. From this study, the authors concluded that disimpaction of canines and moving them to the arch can be done successfully carried out with minimal side effects by means of skeletal anchorage.

Manni et al (2016) conducted a study with the aim of evaluating the effectiveness of the treatment of skeletal Class II malocclusions with an acrylic splint Herbst appliance anchored to miniscrews with 2 types of ligation. Sixty patients (mean age, 11.6 years; SD, 1.9) with a bilateral Angle Class II Division 1 malocclusion were retrospectively selected and divided into 3 homogeneous and balanced groups on the basis of the Herbst anchorage used: without anchorage, miniscrews with elastic chains, and miniscrews with metallic ligatures. A cephalometric sagittal occlusion analysis merged with mandibular incisor proclination and skeletal divergence was carried out before and after treatment. To compare the absolute variations within and among the groups, we performed the 1-sample t test for repeated measures and 1-way analysis of variance, respectively. The results showed overjet was reduced similarly in all groups, the mandibular bone base length increased in the group with elastic chains only, and the change in the distance between Point A and pogonion showed the most reduction in the group with elastic chains ($p < 0.05$). Incisive flaring was more pronounced in the group with no anchorage than in the group with elastic chains ($p < 0.001$) and the group with metallic ligatures ($p = 0.003$). The authors concluded that anchorage to miniscrews with elastic chains increases the orthopedic effect of the acrylic splint Herbst appliance and confirmed that skeletal anchorage reduces incisor flaring.

Turkkahraman et al (2016) conducted a controlled study to evaluate the effects of temporary anchorage devices (TADs) in the treatment of skeletal open bites and to compare the results with untreated controls. A total of forty patients with skeletal anterior open bites were assigned to two groups of twenty each. Titanium miniplates fixed bilaterally to the infrazygomatic crest area were used as TADs and intrusive forces were applied to the posterior teeth with Ni-Ti coil springs. The treatment and normal growth changes were evaluated using 24 measurements (2 angular, 22 linear) with statistically significant differences found between the groups in several of them. In the treatment group, statistically significant upper molar intrusion, posterior rotation of the occlusal plane, anterior rotation of the mandible, and resultant overbite improvement were found. The authors concluded that mild to moderate skeletal anterior open bites could easily be treated with TADs without orthognathic surgery. With the rigid anchorage of miniplates, true molar intrusion of up to 4 mm was achieved. With molar intrusion, anterior rotation of the mandible and a significant reduction in anterior face height were determined.

Lee et al (2015) conducted a clinical study to compare the treatment duration and dentoskeletal changes between two different anchorage systems used to treat maxillary dentoalveolar protrusion and to examine the effectiveness of en-masse retraction using two miniscrews placed in the midpalatal suture. Fifty-seven individuals were divided into two groups according to the method of maxillary posterior anchorage reinforcement: midpalatal miniscrews (25 individuals, mean age 22 years) and conventional anchorage (32 individuals, mean age 19 years). The en-masse retraction period, overall treatment duration, pre-treatment effective ANB angle, and change in the effective ANB angle were compared with an independent samples t-test. Compared to the headgear group, the duration of en-masse retraction was longer by

approximately four months in the miniscrew group ($p < 0.001$). However, there was no significant difference in the total treatment duration between the groups. Moreover, a greater change in the effective ANB angle was observed in individuals treated with miniscrews than in those treated with the conventional method ($p < 0.05$). The authors concluded that treatment using miniscrews placed in the midpalatal area will allow orthodontists more time to control the anterior teeth during en-masse retraction, without increasing the total treatment duration. Furthermore, it achieves better dentoskeletal control than does the conventional anchorage method, thereby improving the quality of the treatment results.

Bechtold et al (2013) conducted a study to determine the effects of linear force vector(s) from interraderic miniscrews on the distalization pattern of the maxillary arch in adults with a Class II malocclusion. Twenty-five adults with mild to moderate Class II dentition and minimal crowding were selected. Either single (group A) or dual (group B) miniscrews were inserted on the posterior interraderic area to deliver a distalizing force to the main archwire. The displacement patterns of maxillary incisors and molars were measured and compared. Significant distalization in the molars and incisors was shown in both groups. Significantly greater distalization and intrusion of the first molar and intrusive displacement of the incisor, together with significant reduction of the mandibular plane, were noted in group B, in contrast to the rotation of the occlusal plane in group A. The authors concluded that interraderic miniscrews predictably induced total arch distalization, leading to the correction of Class II. Additional miniscrews in the premolar area appear to facilitate intrusion and distalization of the entire arch according to the position of the force vectors.

Xun et al (2013) The aim of this retrospective study was to quantitatively evaluate the treatment effects of intrusion of over-erupted maxillary molars using miniscrew implant anchorage and to investigate the apical root resorption after molar intrusion. The subjects included 30 individuals whose average ages were 35.5 ± 9.0 years. All had received intrusion treatments for over-erupted maxillary molars with miniscrew anchorage. There were 38 maxillary first molars and 26 maxillary second molars to be intruded. Two miniscrews were inserted in the buccal and palatal alveolar bone mesial to the over erupted molar. Force of 100-150 g was applied by the elastic chains between screw head and attachment on each side. Lateral cephalograms and panoramic radiographs taken before and after intrusion were used to evaluate dental changes and root resorption of molars. Only six of the 128 miniscrews failed. The first and second molars were significantly intruded by averages of 3.4 mm and 3.1 mm respectively ($p < 0.001$). The average intrusion time was more than 6 months. The crown of the molars mesially tilted by averages of 3.1 degrees and 3.3 degrees ($p < 0.001$) for first and second molars. The amounts of root resorption were 0.2-0.4 mm on average. The intrusion treatment of over erupted molars with miniscrew anchorages could be used as an efficient and reliable method to recover lost restoration space for prosthesis. Radiographically, root resorption of molars was not clinically significant after application of intrusive forces of 200 to 300 g.

Clinical Practice Guidelines

American Association of Oral and Maxillofacial Surgeons (AAOMS)

In a 2023 Parameter of Care, the AAOMS states that mandibular retrusion and maxillary protraction utilizing temporary anchorage devices (TAD) such as screw retained plating systems and miniscrews may be considered as a more conservative, cost efficient, and less morbid treatment alternative to more complex orthognathic surgery for developing skeletal class III malocclusion in pre-and early adolescents.

American Association of Orthodontists (AAO)

In a 2017 clinical practice guideline for orthodontic and dentofacial orthopedics, the AAO states that maxillary and mandibular/dentoalveolar hyper and hypoplasia may require adjunctive anchorage procedures including but not limited to osseointegrated implants, mini-screw implants, miniplates and other temporary anchorage devices.

Corticotomy

A corticotomy performed on the maxilla or mandible is an invasive procedure in which cuts are made in cortical bone to decrease orthodontic treatment time. While the overall body of evidence is large, there is a lack of well-designed studies with larger numbers of participants that show patient selection criteria, and the clinical utility of this procedure. The long-term effects are unknown, and improved orthodontic outcomes and the balance of risks and benefits have not been established.

Zhou et al. (2024) conducted a meta-analysis, to evaluate the safety and efficacy of corticotomy and periodontally accelerated osteogenic (PAOO) interventions, including bone grafting in orthodontic tooth movement. Nineteen studies of 634 individuals were included and the results showed that both interventions significantly reduced orthodontic treatment duration. For corticotomy, in the three studies reporting on canine movement, all three favored the corticotomy group. Gingival indices, plaque indices, and probing depths showed no significant differences between the corticotomy and traditional orthodontic groups. For the PAOO treated groups, four studies showed increased bone thickness, however

there was significant heterogeneity present. The authors concluded that corticotomy can significantly decrease orthodontic treatment duration and accelerate canine movement. In addition, the PAOO procedure reduced the total treatment duration and increased the bone thickness. This meta-analysis is limited by heterogeneity among included studies and lack of long term follow up.

Stefani et al. (2021) conducted a systematic review, to evaluate corticotomy assisted orthodontic treatment (CAOT) as a treatment for expanding narrow arches in adult orthodontic patients. Six studies met the review criteria. The results showed inconsistent results among studies, and that while CAOT can have better results than conventional palatal expansion, it cannot be considered alternative treatment for severe posterior cross bite. Additionally, greater predictability is achieved if CAOT is done in conjunction with bone grafting. The authors concluded that the evidence of corticotomy as a treatment for palatal expansion in adult orthodontic treatment has not been well described in literature and only few published reports are available. Further testing is necessary to confirm the validity of this technique.

Apilaomva et al. (2020) conducted a systematic review to evaluate corticotomies effects on the acceleration or facilitation of tooth movements in different orthodontic treatments. Seven randomized controlled clinical trials and two controlled clinical trials were included in the review. In the selected studies the effectiveness of conventional orthodontic treatment was compared with orthodontics assisted by corticotomy or piezocision. The effect of bone grafts was also evaluated. Variables such as tooth movement, treatment time, bone density and root resorption were studied as well. The methodological quality and evidence of the selected studies was low. Most of the studies observed a statistically significant increase in the rate of dental movement, when performing alveolar corticotomies as an aid in orthodontic treatment; either with the conventional technique or with piezocision. The effect of combining corticotomy with bone grafts was assessed. Corticotomy procedures performed even with conventional methods or piezocision involve a rate increase in dental movement and acceleration during the first months, subsequently returning to baseline values. Bone density may increase as a result of concurrent placement of bone grafting materials during corticotomy procedure. High heterogeneity among studies made it difficult to draw clear conclusions. However, within the limitations of this review, the corticotomy procedures were able to statistically and clinically produce significant temporary decrease in orthodontic tooth movement rate. The available literature about orthodontics facilitated by corticotomy techniques provides low quality evidence, which is why more research is needed. A research with less risk of bias would allow greater comparisons and more significant conclusions.

Gil et al. (2018) conducted a systematic review aimed to provide scientific support to validate alveolar corticotomies as a reliable approach to accelerated orthodontics. Three randomized clinical trials, two prospective randomized clinical trials, six case series and one randomized controlled split-mouth study were included. No clinical trials were retrieved. The results showed the mean total treatment time in corticotomy-facilitated orthodontic cases was 8.85 months; control groups treatment duration was 16.4 months. Complications such as pain, swelling, and dentin hypersensitivity were reported. Few studies mentioned patient/clinician satisfaction. The faster and less invasive procedures appeared to be well tolerated. However, the methodological quality of the selected studies was low, with only low to moderate scientific evidence. The authors concluded that corticotomy-facilitated orthodontics resulted in decreased treatment time. Few complications and low morbidity were found. More solid evidence-based research is required to support these results.

Ji et al. (2017) conducted a study to summarize published systematic reviews (SRs) that assess the effects of adjunctive interventions on the acceleration of orthodontic tooth movement (OTM). Electronic and manual searches were performed up to August 2016. Systematic reviews investigating the impact of adjunctive techniques on the promotion of OTM were included. A total of 11 SRs were included in this study. The results showed the quality of evidence ranged from very low to low. The short-term (one to three months) effects of low-level laser therapy and corticotomy were supported by low-quality evidence. The evidence regarding the efficacy of photo biomodulation, pulsed electromagnetic field, interseptal bone reduction, two vibrational devices (Tooth Masseur and Orthoaccel) and electrical current was of very low quality. Relaxin injections and extracorporeal shock waves were reported to have no impact on OTM according to low- and very low-quality evidence, respectively. Based on currently available information, the authors conclude that low-quality evidence indicates that LLLT and corticotomy are effective to promote OTM in the short term. Future high-quality trials are required to determine the optimal protocols, as well as the long-term effects of LLLT and corticotomy, before warranting recommendations for orthodontics clinics.

Patterson et al. (2016) conducted a systematic review and meta-analysis to examine the evidence for the effectiveness and safety of corticotomy-facilitated orthodontics. Electronic databases were searched for articles that examined the rate of corticotomy-facilitated orthodontic tooth movement and its effects on the periodontium, root resorption, and tooth vitality. Unpublished literature was searched electronically through ClinicalTrials.gov and the ISRCTN registry. Relevant orthodontic journals and reference lists also were checked for eligible studies. Fourteen eligible articles (six RCTs and eight CCTs) were included in this review. The results showed that there was a statistically meaningful increase in the rate of tooth movement compared with controls for all corticotomy techniques assessed. Some studies reported that

acceleration in tooth movement was only temporary (lasting a few months). Corticotomy procedures did not seem to produce unwanted adverse effects on the periodontium, root resorption, and tooth vitality. The quality of the body of evidence was regarded as low owing to the presence of multiple methodologic issues, high risks of bias, and heterogeneity in the included articles. The authors concluded that corticotomy procedures can produce statistically and clinically meaningful temporary increases in the rate of orthodontic tooth movement with minimal side-effects. Additional high-quality randomized clinical trials are needed to allow more definitive conclusions.

Clinical Practice Guidelines

American Association of Oral and Maxillofacial Surgeons (AAOMS)

In a 2023 Parameter of Care, AAOMS indicates that due to the added advantages of lower costs and morbidity, fast recovery, and decreased duration of orthodontic therapy and decompensation, surgically assisted osteogenic orthodontics (SAOO) may be considered as a treatment option when clinically indicated for pediatric patients requiring distraction osteogenesis orthognathic surgery.

U.S. Food and Drug Administration (FDA)

This section is to be used for informational purposes only. FDA approval alone is not a basis for coverage.

Temporary anchorage devices are FDA approved for use in patients aged 12 years and older. There are an extensive number of manufacturers of these devices. Refer to the following website for more information and search by specific product name: <http://www.fda.gov/MedicalDevices/default.htm>. (Accessed June 26, 2025)

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Policy History/Revision Information

Date	Summary of Changes
09/01/2026	Coverage Rationale Corticotomy - Modified content heading; previously named <i>Corticotomy (Not Related to Distraction Osteogenesis or Orthognathic Surgery)</i> - Removed language stating Corticotomy is also known as periodontally accelerated osteogenic orthodontics (PAOO) or surgically assisted osteogenic orthodontics (SAOO) Supporting Information - Updated <i>Description of Services</i>, <i>Clinical Evidence</i>, and <i>References</i> sections to reflect the most current information - Archived previous policy version DCP029.14

Instructions for Use

This Dental Clinical Policy provides assistance in interpreting UnitedHealthcare standard and Medicare Advantage dental plans. When deciding coverage, the member specific benefit plan document must be referenced as the terms of the member specific benefit plan may differ from the standard dental plan. In the event of a conflict, the member specific benefit plan document governs. Before using this policy, check the member specific benefit plan document and any applicable federal or state mandates. UnitedHealthcare reserves the right to modify its policies and guidelines as necessary. This Dental Clinical Policy is provided for informational purposes. It does not constitute medical advice.